Tomoki Nishiyama

https://nishi24yama80.github.io/website/

Education

The University of Chicago — Chicago, USA
2025 - Present

- Real Analysis and Environmental Economics

Graduate School of Economics, The University of Tokyo — Tokyo, Japan
Master's Program in Economics — 2024 - 2025 (on leave)

- First-year PhD sequence in Microeconomics, Macroeconomics, and Econometrics
- Measure Theory, Measure-Theoretic Mathematical Statistics, Measure-Theoretic Stochastic Processes, Graph Theory, and Computational Mathematics

Department of Economics, The University of Tokyo — Tokyo, Japan
B.S. in Economics — 2020 - 2024

Awards and Honors

- **Murata Overseas Scholarship** — 2027.09 - 2029.08
  $50,000 per year for two years
- **Brandt Research Fellowship**, Awarded to an EPIC predoctoral fellow — 2026.09
  Energy Policy Institute at the University of Chicago; $2,500
- **Specially Selected Paper**, Thesis, The University of Tokyo — 2024.03
- **First Prize**, Empirical Micro Conference, Keio University — 2023.07

Working Papers

Bidding Away the Incentives to Drill: Revenue-Sharing Auctions in Mexico's Oil Industry
Studies how the design of revenue-sharing auctions affects bidding behavior and winners' dynamic drilling decisions in Mexico's oil industry, using both reduced-form causal inference and structural estimation. 2026.06 - Present

Production Function Estimation with a Fixed Point Approach
Shows that the production function estimator proposed by Trunschke and Judd (2024) has a fixed-point property, and proposes a new estimation method based on fixed-point iteration that converges 10 times faster than the original GMM estimator. 2026.04 - Present

Work in Progress

Staying for Opportunity: Evidence from an In-Situ Housing Lottery in a Low-Income Community
with Andrew S. Griffen, Yukichi Mano, Yasuyuki Sawada, and Ze Wang

Screening in Public Procurement
with Ze Wang

Patent

Machine Learning Model for Predicting Vacant Houses Using Administrative Data and Water Usage
Japan Patent ID: JP-2024-018823 — 2024.02

Machine Learning Model for Predicting Vacant Houses Using Administrative Data and Utility Consumption Data (Water and Gas)
Japan Patent ID: JP-7137267 — 2022.09

Job Experience

Pre-doctoral Research Professional for Koichiro Ito | Chicago, USA 2025.09 - Present
- Brandt Research Fellow (2026.09 - Present)
- Research Professional at the Energy Policy Institute at the University of Chicago
- Becker Friedman Institute

Research Assistant for Yasuyuki Sawada | Tokyo, Japan 2022.07 - 2025.08
- Conducted six field surveys in Manila, the Philippines.
- Prepared survey questionnaires and managed the surveys with local collaborators in Manila.
- Constructed datasets from the field survey results.
- Analyzed data with causal inference and deep-learning methods.

Research Assistant at Hirofumi Uzawa International Institute | Tokyo and Kyoto, Japan 2023.07 - 2025.08
- Constructed a model integrating environmental economics and macroeconomics.
- Wrote academic articles for the general public.
- Gave academic presentations at Kyoto University.

Research Assistant for Taisuke Nakata | Tokyo, Japan 2021.07 - 2021.12
- Constructed epidemiological-macroeconomic models.
- Ran numerical simulations in MATLAB.
- Prepared policy reports for the Ministry of Health, Labour and Welfare.

GIS Data Analyst at Microbase Inc. | Tokyo, Japan 2020.09 - 2023.07
- Cleaned administrative and utility consumption data.
- Conducted geospatial analysis of administrative data.
- Developed machine learning models to predict vacant houses.
- Optimized the spatial distribution of future energy infrastructure.
- Prepared reports and presentations summarizing research findings.
- Developed web applications.

Teaching Experience

Teaching Assistant for Development Economics | Tokyo, Japan 2025.04 - 2025.07
- Undergraduate course with 200 students.
- Faculty of Economics, The University of Tokyo.

Presentations

- Special Symposium "Evolution of Social Common Capital" (Kyoto University) 2024.03
- Japanese Association for Development Economics, Young Conference (Kyoto University) 2023.11
- Empirical Micro Conference (Keio University) 2023.07
- Korea-Japan Inter University Conference (Online) 2022.11

Skills

Languages: Japanese (Native), English (Fluent), Mandarin Chinese (Intermediate).

Programming: Python, R, Julia, MATLAB, LaTeX, HTML, SQL, GIS (GeoPandas, GDAL, QGIS, PostGIS), Git.